

EDUCATION

University of Waterloo | Computer Engineering, BAsC.

Expected Graduation in April 2028

TECHNICAL SKILLS

Languages & Frameworks: C++ / C (MISRA), Python, Rust, ROS2, RTOS, PLC, Torch, TensorRT, React, Node, Databases

Tools: Docker, Linux, Git, Slurm, CMake, KiCad, Onshape, FEA, Blender, TwinCAT

Hardware: I2C, SPI, CAN, EtherCAT, STM32, ESP32, Jetson, FPGA, Logic Analyzer, Oscilloscope, Soldering

EXPERIENCE

Tesla | Software Engineering Intern, Optimus & HV Integration

Palo Alto, California

Rust, Python, C++/C, Mbed RTOS, Protobuf, ZMQ, CAN, SPI, JS, Beckhoff TwinCAT / EtherCAT

January-April 2026

- Developed the electrical & software architecture for an Optimus tendon reliability tester, integrating linear actuators, loadcell-based feedforward PID in TwinCAT, a python library for test automation, and webserver logging infrastructure, enabling **10x scale-up** of test efforts with seamless integration into existing workflows
- Designed a real-time embedded pyro controller in baremetal C, deriving MCU & ASIC optimization strategies for guaranteed sub-millisecond overvoltage & overcurrent detection & response in existing hardware with **zero impact** to existing system load.
- Implemented async protobuf & ZMQ communications stack in RTOS & Rust for DAQs in a bidirectional HV PSU, improving main controller speed by **50x**

WATonomous Student Design Team | Systems Integration Lead, Self Driving Car

University of Waterloo

C++/C, Python, ROS2, CAN, Torch, CUDA, TensorRT

January 2025 - Present

- Developed a ROS2 car interfacing architecture with CAN and Open Source Car Control, enabling drive-by-wire & real-time odometry with passive safety systems and E-Stop
- Developed end-to-end a Dockerized ROS2 differential drive robot navigation system in sim with costmap generation, map fusion, ray tracing, **A*** path planning and pure pursuit control 
- Migrated YOLOv8-based autonomous car vision pipelines to **TensorRT** resulting in an **inference speedup of 200%**

Independent Robotics | Software Engineering Intern, Navigation and Systems

Montreal, Canada

C++, Python, Docker, ROS2, PX4 Autopilot, Unity, GNSS RTK, AngularJS

May - August 2025

- Architected a Docker monorepo infrastructure and HIL test system for a Jetson & PX4 based autonomous boat, reducing feature development cycles by **40%**.
- Developed containerized **ROS2** nodes for manual and autonomous navigation, long-range serial-to-RF, embedded comms & sensor integration, TCP-Unity sim, and a corresponding Angular.js mission control dashboard.
- Authored a **GNSS RTK driver** with covariance estimation via Jacobian error propagation achieving **sub-degree attitude error**.
- Optimized a polar-to-Cartesian projection driver with Numpy for RTSP sonar, trivializing latency in real-time localization.

Rapyuta Robotics | Software Engineering Intern, Perception and Embedded

Tokyo, Japan

C++/C, Python, Docker, ROS (1), Linux, KiCad, Onshape, Ultralytics, I2C/SPI, UART, ESP32

May - December 2024

- Designed a ROS **TOF** sensor crosstalk-aware obstacle filter FSM, **eliminating** robot collisions for weeks post-deployment.
- Developed a real-time on-device YOLOv8 perception POC, detecting misaligned ASRS bins in test with **95%** accuracy.
- Built end-to-end & deployed a touchscreen ESP32 internal ops device with Wi-Fi/P2P tracking and RFID, I²C BMS interface. Designed PCB, modeled casing in Onshape, soldered and validated hardware, and delivered robust FreeRTOS firmware, earning strong feedback for reliability.
- Developed a unified ROS diagnostic pipeline for all sensors and actuators on the main robot (Jetson) and battery exchange robot (RPI), exposing new error detection for all components and decreasing diagnostics **latency and load by 10x**.

Dept. of Microbiology and Immunology | Cancer Research Laboratory Intern

Western University

Laboratory of Rodney P. DeKoter

August 2021 - November 2022

- Used PCR & restriction enzyme methods to successfully clone EGFP and antibiotic resistance into a plasmid vector.
- Performed Chromatin Immunoprecipitation & Next Generation Sequencing, identifying transcription factor binding sites.
- Published in the Journal of Immunology, securing **\$50,000+** of grant funds. 

PROJECTS

Ross | Arduino, Python, OpenCV, SAM (Meta) | 

Hack the North 2025, University of Waterloo | September 2025

- Built a system that transforms visual art into touch and sound for the blind. Selected as a **Finalist Project** out of **260+** teams.
- Engineered a 3DOF delta plotter using NEMA steppers, servos, 3D-printed parts, & finetuned GRBL firmware for arduino
- Built a pipeline combining Meta's SAM segmentation, graph-based stroke optimization, inverse kinematics and UART, and Cohere+ElevenLabs narration to deliver synchronized tactile and auditory feedback.